VIGNESH

vignesh.k.bala2797@gmail.com | 9600073459 | Chennai.

PROFESSIONAL SUMMARY

Enthusiastic Computer Science graduate with a strong foundation in **data analysis, statistics, and problem-solving**, eager to begin a career as a **Data Analyst or Data Science professional**. A **quick learner and collaborative team player** with hands-on knowledge of **Python, SQL, data cleaning, and data visualization**, passionate about transforming data into **meaningful insights**. Skilled in applying **analytical thinking and proactive approaches** to support **data-driven decision-making, process improvement, and business performance optimization**.

WORK EXPERIENCE

Avasoft Pvt Ltd., *Trainee Engineer.* 10/2025 – 12/2025 | chennai

- Designed and managed **SQL database objects** using **DDL and DML commands** with proper **keys, constraints, and indexing** to ensure **data integrity, scalability, and performance optimization**, along with **data cleaning and preprocessing**.
- Developed **complex SQL queries** using **joins, subqueries, window functions, and aggregations** to perform **exploratory data analysis (EDA)** on large structured datasets.
- Designed **ER diagrams and normalized data models** to support **analytics-ready data pipelines and structured data workflows**.
- Executed **end-to-end ETL workflows**, including **data extraction, transformation, validation, and loading**, using **Azure Data Factory and Azure SQL**.
- Performed **data transformation and processing using Python and PySpark** in **Azure Databricks**, ensuring **high data quality and consistency**.
- Built **interactive dashboards and analytical reports using Power BI**, connecting to **SQL Server and Excel**, and applied **Power Query and Power Pivot** for **data modeling and visualization**.

TECHNICAL SKILLS

Programming & Querying: SQL (Advanced Queries, Joins, Subqueries, Window Functions) | Python (Pandas) | PySpark (Big Data Processing)

Databases & Data Modeling: Data Modeling & Normalization | ER Diagrams | SQL Server | Azure SQL Database | Stored Procedures, Views & Functions

ETL & Cloud Technologies: ETL (Extract, Transform, Load) | Azure Data Factory (ADF) | Azure Databricks | Data Warehousing Concepts

Data Visualization & Reporting: Power BI | Power Query | Power Pivot | Dashboard Development & Reporting

EDUCATION

Masters of science in computer, *Anna University.* 10/2022 – 04/2025 | Chennai

Bachelors of Science in Computer, *Hindustan Collage of Arts and Science, Chennai.* 06/2019 – 05/2022 | Chennai

PROJECTS

BIO DEGRADABLE AND NON BIO DEGRADABLE USING DEEP LEARNING ON CNN ALOGORITHM,
Frontend : HTML , JavaScript , CSS Backend : Python , SQL Database Framework : DJango Tool : VS Code

- Developed a **CNN-based deep learning model** to classify waste images as **biodegradable or non-biodegradable**, supporting **automated waste segregation and sustainable waste management**.
- Implemented and evaluated multiple **CNN architectures** including **Custom CNN (Manual Net), GoogLeNet, MobileNet, and Deploy models** to compare performance and improve **classification accuracy**.
- Collected and curated a **diverse labeled image dataset**, applying **image preprocessing techniques** such as resizing, normalization, and augmentation to enhance model performance.
- Conducted **model training, hyperparameter tuning, and performance evaluation** using accuracy and validation metrics to identify the optimal architecture.
- Designed the solution with a focus on **scalability and deployment readiness** to enable **real-time or automated waste sorting applications**.

ANALYTICS & BUSINESS SKILLS

- Analytical Thinking
- Problem Solving
- Data-Driven Decision Making
- Business Requirement Analysis
- Stakeholder Collaboration

LANGUAGES

English **Tamil** **Telugu**

DECLARATION

I hereby declare that the information provided above is true and accurate to the best of my knowledge and belief.

Vignesh.